Operator Precedence

Operator precedence determines the order in which the operators in an expression are evaluated.

For eg –

*int x = 3 * 4 - 1;*

In the above example, the value of x will be 11, not 9. This happens because the precedence of * operator is higher than - operator. That is why the expression is evaluated as (3 * 4) - 1 and not 3 * (4 - 1).

Operator Precedence Table

Operators	Precedence
postfix increment and decrement	++ --
prefix increment and decrement, and unary	++ -- + - ~ !
multiplicative	* / %
additive	+ -
shift	<< >> >>>
relational	< > <= >= instanceof
equality	== !=
bitwise AND	&
bitwise exclusive OR	^
bitwise inclusive OR	|
logical AND	&&
logical OR	||
ternary	? :
assignment	= += -= *= /= %= &= ^= |= <<= >>= >>>=

Associativity of Operators

If an expression has two operators with similar precedence, the expression is evaluated according to its **associativity** (either left to right, or right to left).

Operators	Precedence	Associativity
postfix increment and decrement	++ --	left to right
prefix increment and decrement, and unary	++ -- + - ~ !	right to left
multiplicative	* / %	left to right
additive	+ -	left to right
shift	<< >> >>>	left to right
relational	< > <= >= instanceof	left to right
equality	== !=	left to right
bitwise AND	&	left to right
bitwise exclusive OR	^	left to right
bitwise inclusive OR	|	left to right
logical AND	&&	left to right
logical OR	||	left to right
ternary	? :	right to left
assignment	= += -= *= /= %= &= ^= |= <<= >>= >>>=	right to left

Note - These notes are just for a quick glance. We don't have to memorize them all at once. Most of these rules are very logical and we have been following them in a lot of instances already.